

Abhishek Karna

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Research Interests: AI for Quantum Materials; Theoretical Quantum Condensed Matter Physics

EDUCATION

- Princeton University** Princeton, NJ
PhD in Physics; Advisor: Prof. B. Andrei Bernevig Aug. 2026 – 2031
- Duke University** Durham, NC
BS Physics, BS Mathematics, Certificate in Philosophy, Politics, and Economics (PPE) Aug. 2022 – May 2026
Advisors: Profs. Shailesh Chandrasekharan, Sara Haravifard, Brian G. McAdoo

PUBLICATIONS

- Karna, A.**, Wu, H. S., Chandrasekharan, S., and Kaul, R. K. *Projected Density Matrix Sampling for Lattice Hamiltonians*. *Physical Review D* 113, 074511 (2026). [PRD](#) [arXiv:2511.19209](#)
- Karna, A.** *Theoretical and Experimental Investigations of Novel Quantum Magnets*. Duke University Honors Thesis (2026; defense passed).
- Shrestha, H.*, **Karna, A.***, et al. *Investigation of the cross-border movement of PM_{2.5} in the Terai Belt of Nepal using a low-cost sensor network*. *Atmospheric Pollution Research* (2026). [doi:10.1016/j.apr.2026.102920](#) (*Co-first author)

RESEARCH EXPERIENCE

- AI + Quantum Materials** Princeton, NJ
Supervisor: Prof. B. Andrei Bernevig, Princeton University Apr. 2026 – Present
Developing data-driven methods to predict and discover quantum materials properties (magnetic order, superconducting transition temperatures, and synthesizability) that are difficult to access from first principles alone.
- Frustrated Quantum Spin Systems** Durham, NC
Supervisor: Prof. Shailesh Chandrasekharan, Duke University May 2024 – May 2026
Developed the *Projected Density Matrix Sampling (PDMS)* method, a new continuous-time quantum Monte Carlo algorithm free of Trotter errors, benchmarked on the Shastry-Sutherland model from 2 to 96 sites. [Paper](#) [Presentation](#) [Poster](#)
- Quantum Materials Synthesis and Characterization** Durham, NC
Supervisor: Prof. Sara Haravifard, Duke University Aug. 2022 – May 2026
Solid-state synthesis and arc melting of frustrated spinels (ZnAl₂O₄, ZnV₂O₄), borides (TmB₄, HoB₄), and perovskites (NdFeO₃), including optical floating zone single-crystal growth. Structural characterization via high-resolution XRD with Rietveld refinement (FullProf), Laue scattering, and PPMS magnetometry. [Poster 1](#) [Poster 2](#) [Poster 3](#)
- Atmospheric Pollution and Climate Physics** Durham, NC
Supervisor: Prof. Brian G. McAdoo, Duke University Nov. 2022 – May 2026
Applied HYSPLIT back-trajectory modeling, aerosol optical depth analysis, and PSCF to identify transboundary PM_{2.5} transport pathways in Nepal and the Indo-Gangetic Plain. Analyzed diurnal patterns, planetary boundary layer height, and ventilation coefficients to isolate local vs. non-local pollution sources. [Presentation](#)
- Collider Physics for HL-LHC Upgrade** Durham, NC
Supervisor: Prof. Ashutosh Kotwal, Duke University Apr. 2023 – Aug. 2023
Designed wedge-generation and spacepoint-divider algorithms for parallelized silicon track-trigger reconstruction targeting metastable dark matter candidates at the HL-LHC. [Poster](#)

RESEARCH PRESENTATIONS AND CONFERENCES

- APS Global Physics Summit 2026 – Combined March & April Meeting** Denver, Colorado
Projected Thermal Density Matrix Monte Carlo on the Shastry-Sutherland Model Mar. 16–20, 2026
- International Conference on Strongly Correlated Quantum Materials (ICSCQM)** Hvar, Croatia
Low-energy spectrum of the Shastry-Sutherland Model on small lattices using a new Monte Carlo Oct. 12–17, 2025
- APS Global Physics Summit 2025 – Combined March & April Meeting** Anaheim, California
Experimental synthesis, single crystal growth, & characterization of NdFeO₃ Mar. 16–22, 2025

HONORS AND AWARDS

- Karsh International Scholarship, Duke University (2022)**: Full-ride merit scholarship (\$350,000+) for exceptional interdisciplinary research and civic engagement. [View Award](#)
- PSI Start, Perimeter Institute for Theoretical Physics (2025)**: Competitive 10-week program covering group theory, supersymmetric quantum mechanics, tensor networks, and cosmic structure formation. [View Program](#)
- Dean's Summer Research Fellowship (DSRF), Duke University (2025)**: Fellowship supporting theoretical QMC research on the Shastry-Sutherland model. [View Award](#)
- First Prize, MIT Interdisciplinary Quantum Hackathon iQuHack (2025)**: Quantinuum challenge; simulation of open and closed quantum systems via Hamiltonians and Lindbladians. [View Project](#)
- Huang Fellow, Duke Initiative for Science and Society (2023)**: Selective program (20 scholars) integrating science and liberal arts leadership; fully funded summer research. [View Award](#)
- Global Awardee, Beamline for Schools, CERN (2021)**: Awarded for a research proposal on targeted cancer treatment using Delta rays. [View Award](#)

TEACHING

- Teaching Assistant: Quantum Mechanics I (Physics 464), Duke University**: Fall 2025; assisted Prof. Socolar in course instruction; held weekly problem-set discussions and graded assignments and exams.

SERVICE AND LEADERSHIP

- Duke Society of Physics Students (SPS)**: Co-President (2024–25) and Event Coordinator (2023–24); organized a Physics Research Seminar Series and Alumni Seminar Series; fostered community and networking within the Duke Physics department.